BRAC UNIVERSITY

Inspiring Excellence

CSE370 : Database Systems

Project Report

Project Title : AI-Powered Mental Health Tracker

Group No : 8	CSE370 Lab Section : 15	Summer 2025
ID	**Name**	**Contribution**
23101499	Maisha Maisara	Frontend and User Interface
23301073	Lamia Hai Meghla	AI Integration
22201350	Sumaiya Rahman	Backend and Database and styling

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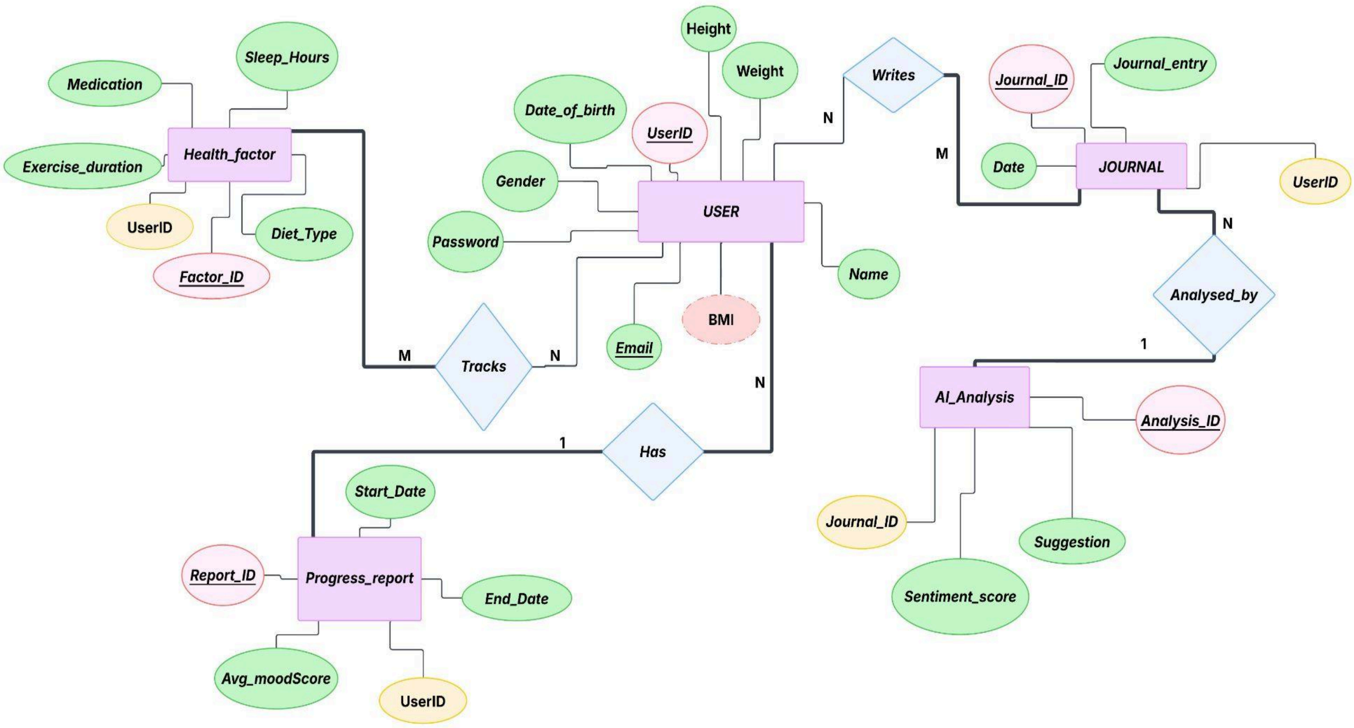
Introduction

The AI-Powered Mental Health Tracker is a website-based application that enables users to track and improve their mental health. Users can sign up or log in to the system and then access features like health factor tracking, progress reporting, journaling, and keeping track of their personal profiles. Users can enter daily mood ratings on a simple 1-10 scale, which are stored in the database and also shown in the website. Users can also write journals, which are examined using Hugging Face's sentiment analysis engine to provide a sentiment score and an automated suggestion based on the score. The device also saves health-related data such as sleep, exercise, diet, and medications for future reference. BMI is also calculated using the user's height and weight.

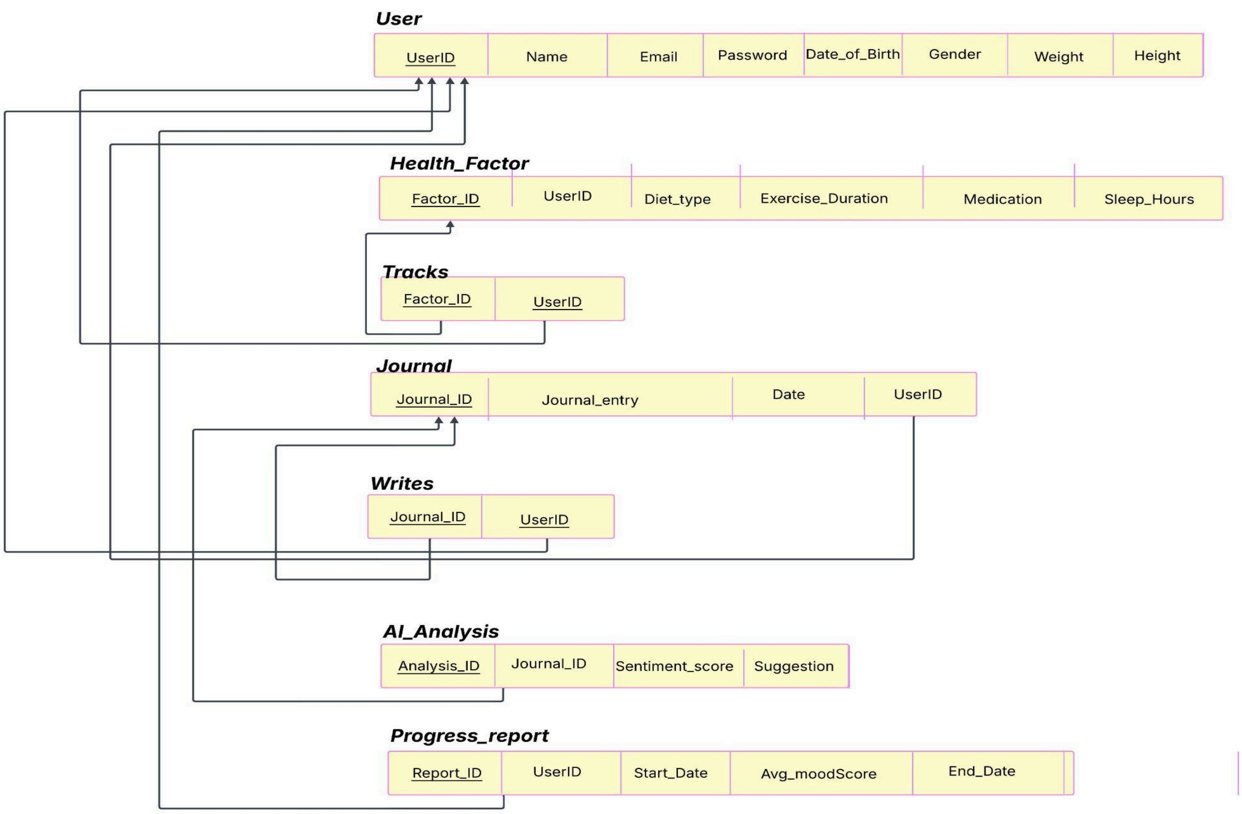
Project Features

ID, Name	Features [3 per member]	
23101499, Maisha Maisara	Ft 1	Designed and implemented all user interfaces (login, signup, journal, health factors, progress report, profile).
	Ft 2	Integrated form validation and user-friendly inputs (mood slider, journal text, health factor fields).
	Ft 3	Added BMI in the profile option which is calculated with the help of height and weight.
23301073, Lamia Hai Meghla	Ft 1	Integrated AI-based sentiment analysis for journal entries to give a sentiment score.
	Ft 2	Connected AI results to give an automatic suggestion.
	Ft 3	Created a token and selected AI from the hugging face platform.
22201350, Sumaiya Rahman	Ft 1	Designed and maintained MySQL database schema with constraints and relationships.
	Ft 2	Implemented backend logic for data insertion, updates, and auto-handling of End_Date in progress reports.
	Ft 3	Managed database connectivity with PHP (DBconnect.php) and implemented consistent navigation (Back to Home button, page layouts, styling).

ER/EER Diagram



Schema Diagram



Normalization

a. Explain if your converted Schema is in 1NF or not. If not, decompose it to 1NF.

Ans: It is in 1NF, because there are no multivalued/composite attributes or nested relations present in the schema.

b. Explain if your converted Schema is in 2NF or not. If not, decompose it to 2NF. Can there be any partial functional dependencies in your relational schema?

Ans: It is in 2NF already because there is no partial dependency in the schema.
But there can be a partial functional dependency in “Tracks”. Because in Tracks, Factor_ID came from “Health factor" and User_ID came from “User”.
Also in “Writes” the Journal_ID came from Journal and User_ID came from “User”.

c. Explain if your converted Schema is in 3NF or not. If not, decompose it to 3NF. Can there be any transitive dependencies in your relational schema?

Ans: It is also in 3NF because there is no Transitive dependency. In our relational schema there aren’t any transitive dependencies.

Frontend Development

The frontend of the AI-Powered Mental Health Tracker is built using HTML and CSS to give a clean, intuitive, and user-friendly interface. It features important areas including login, sign up, homepage, health factor management, progress reporting, journal writing, profile settings and AI analyzing. The design prioritizes simplicity and clarity, minimizing needless complexity while allowing users to effortlessly navigate between features. Interactive components, such as the daily mood scale, are used to facilitate user input. The frontend connects with the backend via form submissions and server responses, ensuring that data input by users is saved, evaluated, and represented in reports.

Backend Development

The backend of the AI-Powered Mental Health Tracker is written in PHP. A MySQL database maintains user data, health indicators, journals, progress report, while derived properties like BMI are calculated dynamically. Journals are integrated with the Hugging Face API for sentiment analysis, allowing AI-driven score and automated suggestion, but mood monitoring using the 1-10 scale is handled directly in the backend without AI. PHP sessions manage user authentication and activity tracking to ensure secure database interactions.

Source Code Repository

https://github.com/maisha-maisara/CSE370

Project files: https://github.com/maisha-maisara/CSE370/tree/project

Conclusion

The AI-Powered Mental Health Tracker is an user-friendly application that helps people keep track of their mental health. It delivers individualized insights by combining daily mood tracking, optional AI-driven journal analysis, and health factor reporting. Daily mood scales promote consistent interaction, while a refined frontend and well-organized backend enable easy navigation. Overall, the application shows how technology can effectively promote mental health awareness and self-care.

References

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